

Entity Sets in DBMS

What is an Entity?

- An entity is a real-world object that can be identified.
- It can be a person, place, object, or event.
- Entities store information in databases.

Examples

- Student
- Employee
- Book
- Course

What is an Entity Set?

- A collection of similar entities is called an Entity Set.
- All entities in the set have similar attributes.

Example

Student Entity Set

Student_ID	Name	Course
101	Ravi	B.Tech
102	Anu	B.Sc

- All student records together form the Student Entity Set.

Important Points About Entity Sets

Basic Understanding

1. Entity set is a group of similar entities.
2. Entities in the set share common properties.
3. Entity sets are represented as rectangles in ER diagrams.
4. Entity sets help organize database information.
5. Each entity in the set is unique.
6. Entity sets contain attributes.
7. Entity sets are important in database design.

8. Entity sets help store related data together.
9. Entity sets make data management easier.
10. Every database contains multiple entity sets.

Examples of Entity Sets

Entity Set	Example Entities
Student	Ravi, Anu, Kiran
Employee	Ram, Sita, John
Book	DBMS Book, Java Book
Course	CSE, ECE, MBA

Types of Entity Sets

1. Strong Entity Set

Definition

- An entity set that has a primary key.
- It can exist independently.

Characteristics

11. Has its own primary key.
12. Does not depend on other entities.
13. Represented by a single rectangle.

Example

- Student
- Employee

Example Attributes

- Student_ID
- Name
- Age

2. Weak Entity Set

Definition

- An entity set that does not have a primary key.
- Depends on another entity set.

Characteristics

14. Cannot exist independently.
15. Depends on strong entity set.
16. Represented by double rectangles.
17. Uses partial key.

Example

- Dependent of Employee
- Order Items

Entity vs Entity Set

Entity	Entity Set
Single object	Collection of objects
Example: Ravi	Example: Students
One record	Group of records
Represents one instance	Represents many instances

Characteristics of Entity Sets

18. Entity sets contain entities.
19. Entities have attributes.
20. Entity sets may have relationships.
21. Entity sets are used in ER models.
22. Each entity set has a name.
23. Entity sets help reduce redundancy.
24. Entity sets improve database organization.
25. Entity sets support efficient searching.
26. Entity sets help maintain consistency.

27.Entity sets may contain key attributes.

Key Attributes in Entity Sets

28.Key attribute uniquely identifies an entity.

29.Primary key is very important.

30.No duplicate primary key values allowed.

Example

- Student_ID in Student entity set

ER Diagram Representation

Strong Entity Set

```
+-----+  
| STUDENT |  
+-----+
```

Weak Entity Set

```
|| DEPENDENT ||
```

Advantages of Entity Sets

- 31.Easy data organization
- 32.Reduces duplication
- 33.Improves data management
- 34.Helps in relationship mapping
- 35.Makes database design simple
- 36.Supports efficient storage
- 37.Easy to understand
- 38.Useful for large databases
- 39.Helps maintain integrity
- 40.Simplifies querying

Additional Important Points

- 41.Entity sets are part of ER model.
- 42.Entity sets can participate in relationships.
- 43.One entity set can connect with another.

- 44.Entity sets contain multiple records.
- 45.Strong entity sets are independent.
- 46.Weak entity sets are dependent.
- 47.Entity sets are converted into tables.
- 48.Each row in table represents an entity.
- 49.Entity sets help represent real-world systems.
- 50.Proper entity set design improves database performance.

Real-Time Example

College Database

Entity Sets

- Student
- Faculty
- Course
- Department

Relationships

- Student enrolls in Course
- Faculty teaches Course

Short Definition for Exams

Entity Set:

A collection of similar entities having the same attributes is called an entity set.

One-Mark Important Points

- Entity set is a group of similar entities.
- Strong entity set has primary key.
- Weak entity set depends on strong entity set.
- Entity sets are represented by rectangles in ER diagrams.
- Entity sets contain attributes and relationships.